Auto-immune polyradiculoneuropathy and a novel IgG biomarker in workers exposed to aerosolized porcine brain.

Twenty-four patients, all of whom were exposed to aerosolized porcine brain tissue through work-place environment (abattoir), developed a syndrome of immune-mediated polyradiculoneuropathy; three also had central nervous system manifestations (transverse myelitis, meningoencephalitis, and aseptic meningitis). Patients had characteristic electrophysiological findings of very distal and proximal conduction slowing (prolonged distal and F-wave latencies, regions where the blood-nerve barrier is the most permeable) and all patients' serum contained a novel IgG immunofluorescence pattern. Nerve pathology, when available, showed mild changes of segmental demyelination, axonal degeneration, and inflammatory changes. Patients had meaningful improvement of symptoms and electrophysiologic findings with immune therapy and with removal of exposure to aerosolized brain tissue. We postulate that this outbreak is an auto-immune polyradiculoneuropathy triggered by occupational exposure to multiple aerosolized porcine neural tissue antigens that result in neural damage where the blood-nerve barrier is the least robust.